

A “Politically Robust” Experimental Design for Public Policy Evaluation, with Application to the Mexican Universal Health Insurance Program

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Joint work with Emmanuela Gakidou, Nirmala Ravishankar, Ryan T. Moore, Jason Lakin, Manett Vargas, Martha María Téllez-Rojo, Juan Eugenio Hernández Ávila, Mauricio Hernández Ávila, Héctor Hernández Llamas

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Before Treatment



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After Treatment



(Manett's) Arturo Vargas

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- Our plan: **fail-safe research design components**

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 - **Politically Infeasible**: local officials want benefits for their favored areas first

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- If decisions are arbitrary, they can be randomized

Is Randomization Always Unethical?

- Not ethical to randomly assign health care to Mexicans
- Is it ok to randomly assign whether people are told on the left or right side of the road first?
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- **Generalization: randomization is acceptable at one level below that at which politicians care**

A Feasible Design for Scientific Evaluation

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 - **Methodological**: Drop areas where affiliation had already started

Remaining in study: 148 clusters in 7 states



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- No randomization at individual level
- Without an evaluation, choices would still be made, but would be arbitrary choices made by local government officials

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 - Consequence: Bias in evaluation conclusions
- We need estimators robust not merely to statistical assumptions but to real world problems

We Use: Paired Matching, then Randomization

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Experimental Design Implementation

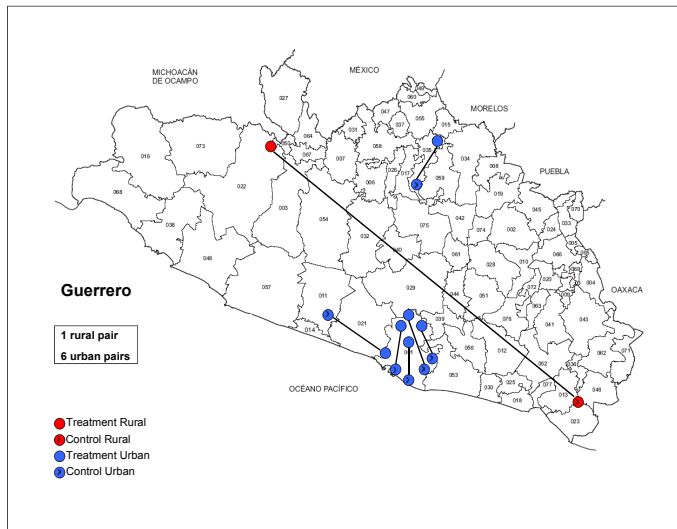
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Experimental Design Implementation

- At the last moment: Flip coin to choose treatment and control cluster for each pair
- Treatment assignments delivered to state governments
- Intensive affiliation begins in treatment clusters
- 74 matched treatment-control pairs in the evaluation: 55 rural and 19 urban in 7 states

State	Rural Pairs	Urban Pairs	Total
Guerrero	1	6	7
Jalisco	0	1	1
México	35	1	36
Morelos	12	9	21
Oaxaca	3	1	4
San Luis Potosí	2	0	2
Sonora	2	1	3
<i>Total</i>	<i>55</i>	<i>19</i>	<i>74</i>

Matched Pairs, Guerrero

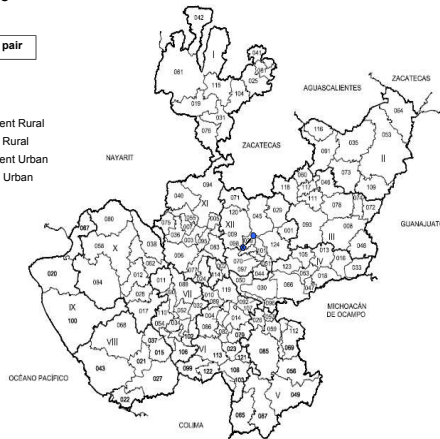


Matched Pairs, Jalisco

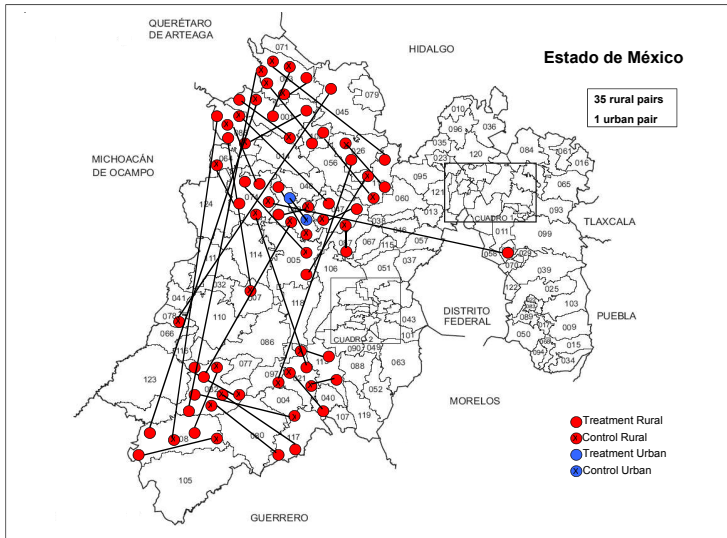
Jalisco

1 urban pair

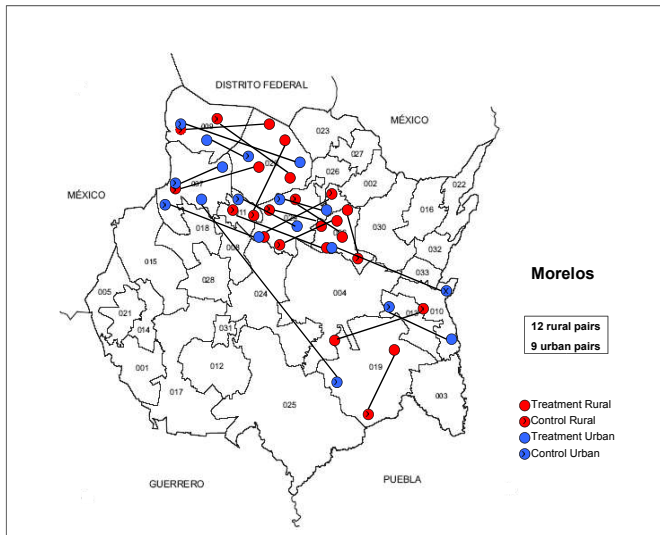
- Treatment Rural
- Control Rural
- Treatment Urban
- Control Urban



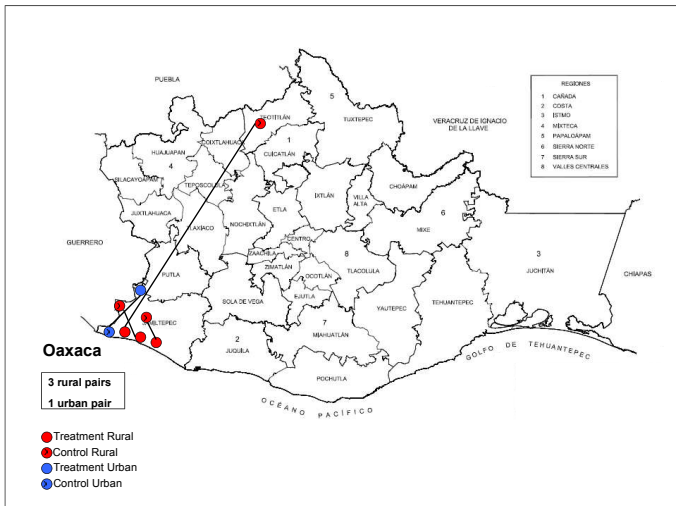
Matched Pairs, Estado de México



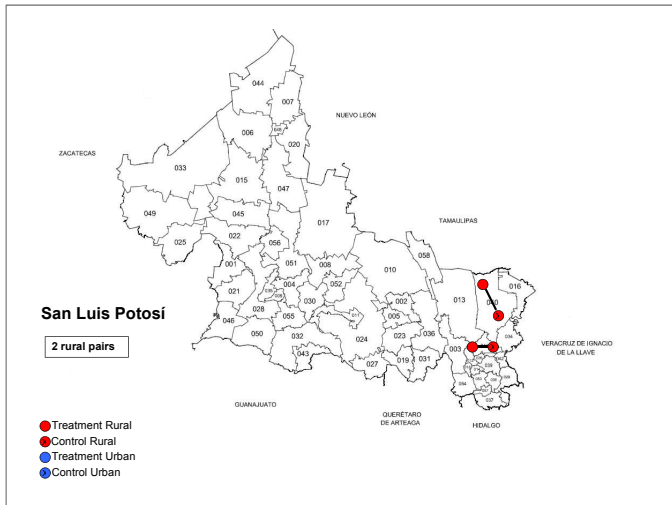
Matched Pairs, Morelos



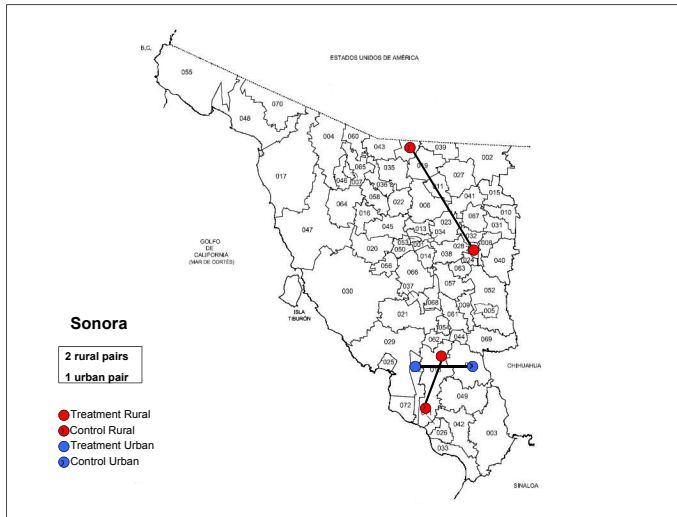
Matched Pairs, Oaxaca



Matched Pairs, San Luis Potosí



Matched Pairs, Sonora



Evaluation Design is Triply Robust

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Design has three parts

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If matching **or** randomization **or** parametric analysis is right, but the other two are wrong, results are still unbiased

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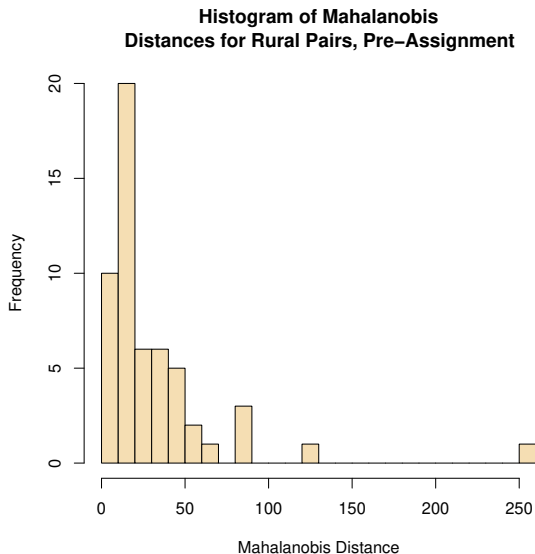
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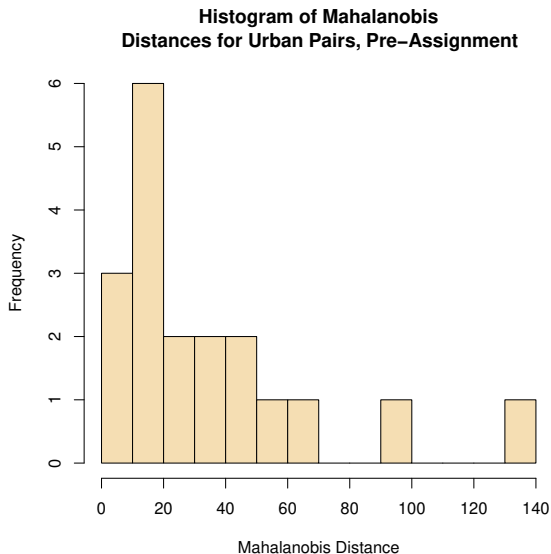
Two Additional Checks if Triple Robustness Fails

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- 2 If we lose pairs, we check for selection bias by rerunning this check

Total Multivariate Distances Within All 55 Rural Pairs

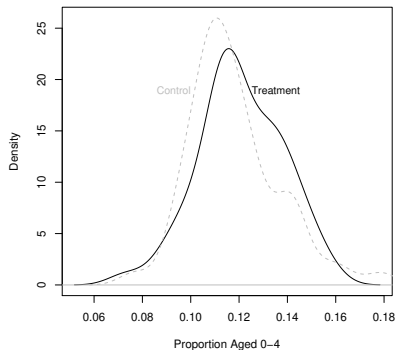


Total Multivariate Distances within All 19 Urban Pairs

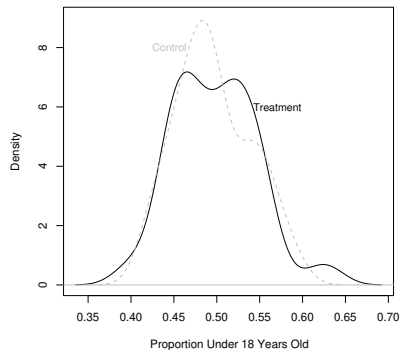


Rural Age Balance After Randomization

Smoothed Histogram of Proportion Aged 0–4, Rural Clusters, Post-Assignment

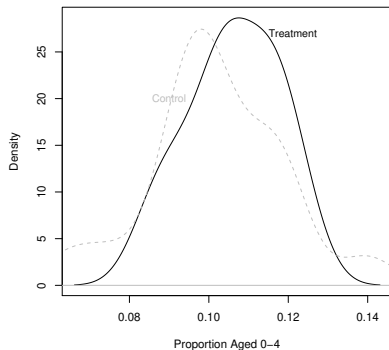


Smoothed Histogram of Proportion Under 18 Years Old, Rural Clusters, Post-Assignment

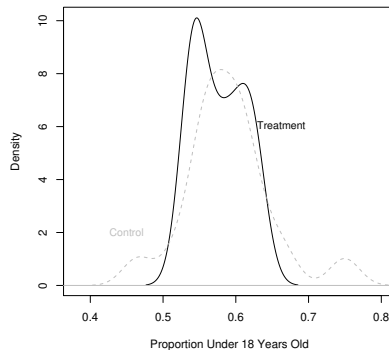


Urban Age Balance After Randomization

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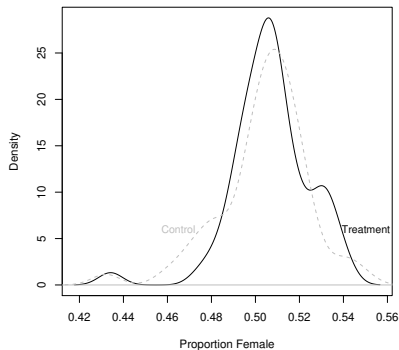


Smoothed Histogram of Proportion Under 18 Years Old, Urban Clusters Post-Assignment

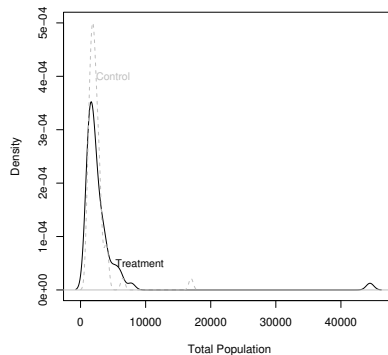


Rural Demographic Balance After Randomization

Smoothed Histogram of Proportion Female, Rural Clusters, Post-Assignment

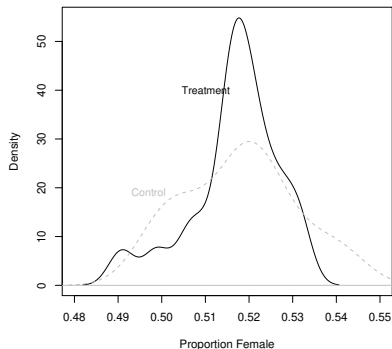


Smoothed Histogram of Total Population, Rural Clusters, Post-Assignment

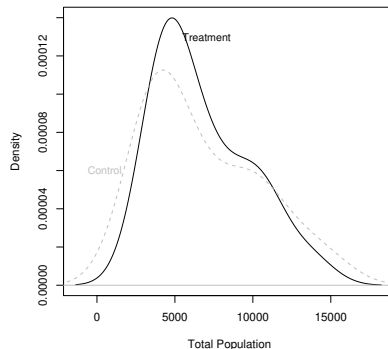


Urban Demographic Balance After Randomization

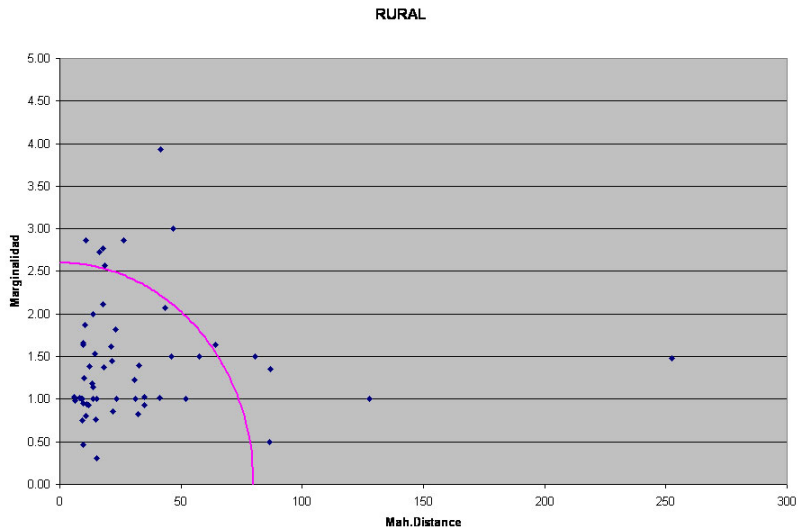
Smoothed Histogram of Proportion Female, Urban Clusters, Post-Assignment



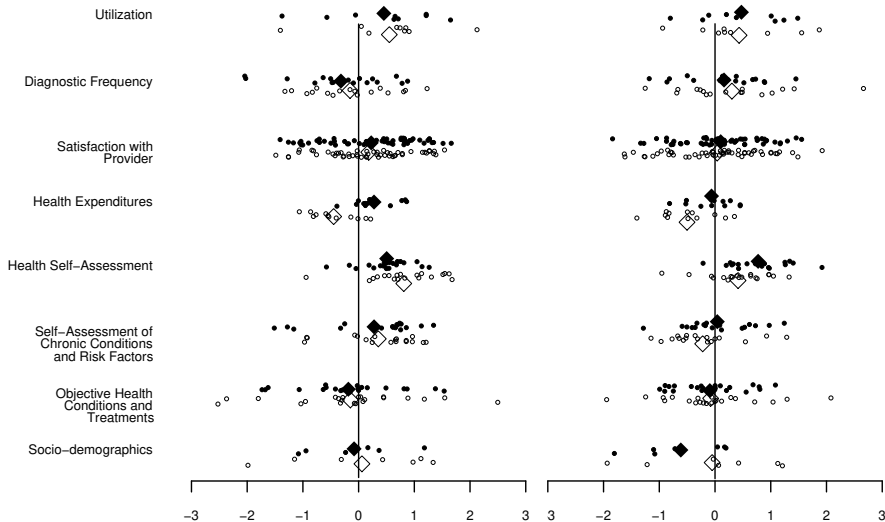
Smoothed Histogram of Total Population, Urban Clusters, Post-Assignment



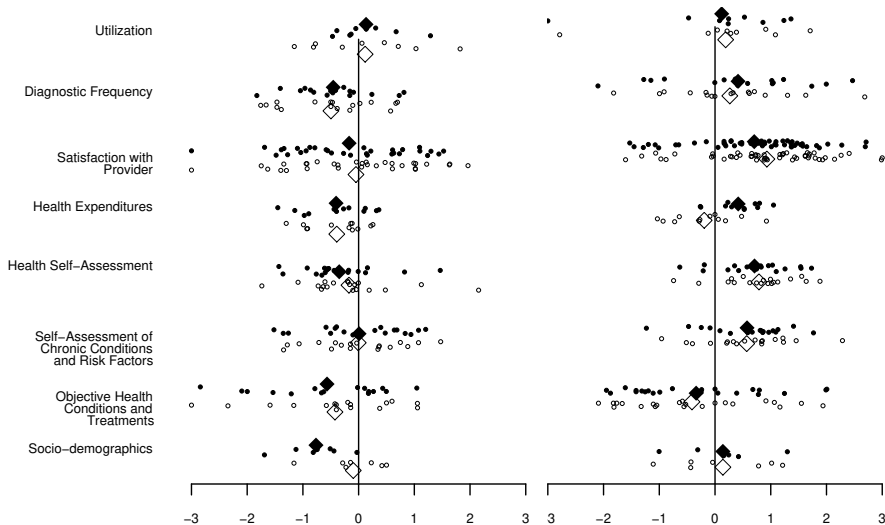
Choosing Pairs for the Survey



ITT on Outcome Measures at Baseline, for all families (left) and poor families, in Oportunidades (right)

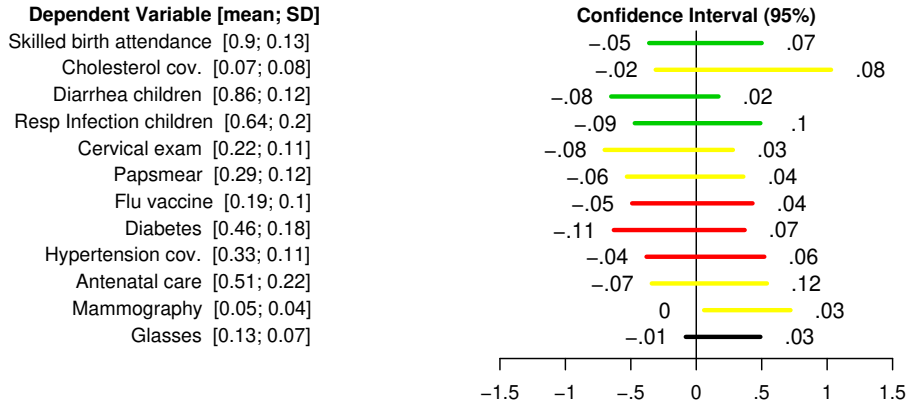


ITT on Outcome Measures at Baseline, for wealthy families (left) and middle income families (right)



Effect of SP Rollout at Baseline: 1 of many

(Expected effects at 10 months: **small**, **medium**, **large**)



For more information

<http://GKing.Harvard.edu>

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- Exact match on state and urban/rural

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